Modeling Neutron Stars
Lecture-5

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Contents:

Elementary Field Theory

- Recap
- Dirac Equation
- Plane wave solution of Dirac Equation
- Lagrangian for Dirac Field

Dirac Equation

γ matrices

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- The commutation relation between γ^μ and $\sigma^{\mu\nu}$ was established as,

$$\boxed{[\gamma_\tau, \sigma_{\mu\nu}]_- = 2i\left(g_{\tau\mu}\gamma_\nu - g_{\tau\nu}\gamma_\mu\right)}$$

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- This eventually led us to

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Angular momentum in Dirac field

Spinors

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\Rightarrow J_{\mu\nu} &= \frac{1}{2}\sigma_{\mu\nu} - i(x_\nu \partial_\mu - x_\mu \partial_\nu)
\end{aligned}$$

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\end{aligned}$$

- The finite transformation is given by exponentiating the infinitesimal transformation as

$$\psi'(x') = S(\Lambda)\psi(x) = \exp\left(-\frac{i}{4}\sigma_{\mu\nu}\omega^{\mu\nu}\right)\psi(x).$$

Lorentz Transformation:

scalar, vector, tensor

- **Problem 1: Show that under Lorentz transformation $\overline{\psi}\gamma^{\mu}\psi$ transforms as a vector and $\overline{\psi}\sigma^{\mu\nu}\psi$ transforms as tensor, where, $\overline{\psi} \equiv \psi^{\dagger}\gamma^{0}$.**

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Remember, γ^μ is not a vector!!!!!

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$$\Rightarrow \quad \overline{\psi}S^{-1}\sigma^{\mu\nu}S\psi = \frac{i}{2}\overline{\psi}S^{-1}\left(\gamma^{\mu}\gamma^{\nu} - \gamma^{\nu}\gamma^{\mu}\right)S\psi$$

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$$= \frac{i}{2}\overline{\psi}\left[\left(S^{-1}\gamma^{\mu}S\right)\left(S^{-1}\gamma^{\nu}S\right) - \left(S^{-1}\gamma^{\nu}S\right)\left(S^{-1}\gamma^{\mu}S\right)\right]\psi$$

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&= \frac{i}{2}\overline{\psi}S^{-1}\left(\gamma^{\mu}SS^{-1}\gamma^{\nu}-\gamma^{\nu}SS^{-1}\gamma^{\mu}\right)S\psi \\
&= \frac{i}{2}\overline{\psi}\left[\left(S^{-1}\gamma^{\mu}S\right)\left(S^{-1}\gamma^{\nu}S\right)-\left(S^{-1}\gamma^{\nu}S\right)\left(S^{-1}\gamma^{\mu}S\right)\right]\psi \\
&= \frac{i}{2}\overline{\psi}\left[\left(\Lambda^{\mu}{}_{\rho}\gamma^{\rho}\right)\left(\Lambda^{\nu}{}_{\tau}\gamma^{\tau}\right)-\left(\Lambda^{\nu}{}_{\tau}\gamma^{\tau}\right)\left(\Lambda^{\mu}{}_{\rho}\gamma^{\rho}\right)\right]\psi
\end{aligned}$$

Lorentz Transformation:

scalar, vector, tensor

- Similarly under Lorentz transformation

$$
\begin{aligned}
& \overline{\psi}\sigma^{\mu\nu}\psi \xrightarrow{\text{L.T}} \overline{\psi}S^{-1}\sigma^{\mu\nu}S\psi \\
\Rightarrow \quad & \overline{\psi}S^{-1}\sigma^{\mu\nu}S\psi = \frac{i}{2}\overline{\psi}S^{-1}\left(\gamma^{\mu}\gamma^{\nu} - \gamma^{\nu}\gamma^{\mu}\right)S\psi \\
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& = \overline{\psi}\Lambda^{\mu}{}_{\rho}\Lambda^{\nu}{}_{\tau}\left\{\frac{i}{2}\left[\gamma^{\rho}\gamma^{\tau} - \gamma^{\tau}\gamma^{\rho}\right]\right\}\psi
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Similarly, a pseudoscalar and pseudovector transform as $A \xrightarrow{\text{L.T}} \det|\Lambda|A$ and $A^{\mu} \xrightarrow{\text{L.T}} \det|\Lambda|\Lambda^{\mu}{}_{\nu}A^{\nu}$.

Problems:

- **Problem 2: Show that under Lorentz transformation $\overline{\psi}\psi$ transforms as a scalar, $\overline{\psi}\gamma_5\psi$ transforms as a pseudoscalar and $\overline{\psi}\gamma_5\gamma^\mu\psi$ transforms as a pseudovector.**

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- **Problem 2: Show that under Lorentz transformation $\overline{\psi}\psi$ transforms as a scalar, $\overline{\psi}\gamma_5\psi$ transforms as a pseudoscalar and $\overline{\psi}\gamma_5\gamma^\mu\psi$ transforms as a pseudovector.**
- **Problem 3: Prove the following contraction formulae:**

$$\begin{aligned}
\gamma_\lambda\gamma^\lambda &= 4, \\
\gamma_\lambda\gamma_\mu\gamma^\lambda &= -2\gamma_\mu, \\
\gamma_\lambda\gamma_\mu\gamma_\nu\gamma^\lambda &= 4\boldsymbol{g}_{\mu\nu}, \\
\gamma_\lambda\gamma_\mu\gamma_\nu\gamma_\rho\gamma^\lambda &= -2\gamma_\rho\gamma_\nu\gamma_\mu.
\end{aligned}$$

NB: These are matrix relations!!!

Solution of Dirac Equation

Single particle interpretation

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- For a general value of 3-momentum $\boldsymbol{p}$, one can represent the solution as

$$\psi(x) = \begin{cases} u_s(\boldsymbol{p})e^{-ip\cdot x}, \text{ with, } \left(\sqrt{\boldsymbol{p}^2 + m^2}, \boldsymbol{p}\right) \\ v_s(\boldsymbol{p})e^{+ip\cdot x}, \text{ with, } \left(-\sqrt{\boldsymbol{p}^2 + m^2}, -\boldsymbol{p}\right) \end{cases}$$

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- So we obtain by putting these in Dirac equation

$$\begin{aligned} \left(\gamma^\mu p_\mu - m\right) u_s(\boldsymbol{p}) &= 0 \\ \left(\gamma^\mu p_\mu + m\right) v_s(\boldsymbol{p}) &= 0 \end{aligned}$$

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- Using the Dirac conjugation of ψ defined as $\overline{\psi} \equiv \psi^\dagger \gamma^0$, one can rewrite the Dirac equation for u_s and v_s as

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\end{aligned}
$$

Similarly, one can write,

$$
\overline{v}_s(\boldsymbol{p}) \left(\gamma^\mu p_\mu + m\right) = 0
$$

Gordon Identity

Normalization for spinors

- Using the Dirac equation for u_s and its Hermitian conjugate,

$$\left(\gamma^{\nu} p_{\nu} - m\right) u(\boldsymbol{p}) = 0, \quad \text{and}, \quad \overline{u}(\boldsymbol{p}')\left(\gamma^{\nu} p'_{\nu} - m\right) = 0$$

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Combining them together we can write

$$\overline{u}(\boldsymbol{p}')\left(\gamma^\mu\gamma^\nu p_\nu + \gamma^\nu\gamma^\mu p'_\nu\right) u(\boldsymbol{p}) = 2m\overline{u}(\boldsymbol{p}')\gamma^\mu u(\boldsymbol{p})$$

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\Rightarrow \quad & \overline{u}(\boldsymbol{p}')\left(\frac{1}{2}\left[\gamma^\mu,\gamma^\nu\right]_+ p_\nu + \frac{1}{2}\left[\gamma^\mu,\gamma^\nu\right]_+ p'_\nu + \frac{1}{2}\left[\gamma^\mu,\gamma^\nu\right]_- p_\nu\right. \\
& \left. - \frac{1}{2}\left[\gamma^\mu,\gamma^\nu\right]_- p'_\nu\right) u(\boldsymbol{p}) = 2m\overline{u}(\boldsymbol{p}')\gamma^\mu u(\boldsymbol{p})
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\end{aligned}$$

Normalization

Problems

- **Problem 4: Find the Gordon Identity for v-spinors. Show that combining that with the identity of u-spinors one can write the normalization conditions as**

$$\overline{u}_r(p)u_s(p) = -\overline{v}_r(p)v_s(p) = 2m\delta_{rs}.$$

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- The normalization conditions are given by,

$$u_r^\dagger(\boldsymbol{p})u_s(\boldsymbol{p}) = v_r^\dagger(\boldsymbol{p})v_s(\boldsymbol{p}) = 2E_p\delta_{rs},$$
$$v_r^\dagger(\boldsymbol{p})u_s(-\boldsymbol{p}) = u_r^\dagger(\boldsymbol{p})v_s(-\boldsymbol{p}) = 0.$$

Solution of Dirac Equation

Dirac-Pauli Representation

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- In Dirac-Pauli representation the γ matrices are given by,

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix},$$

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- Putting them in Dirac equation for u-spinors

$$\gamma^\mu p_\mu - m = \gamma^0 E_p - \boldsymbol{\gamma} \cdot \boldsymbol{p} - m = \begin{pmatrix} E_p - m & -\boldsymbol{\sigma} \cdot \boldsymbol{p} \\ \boldsymbol{\sigma} \cdot \boldsymbol{p} & -E_p - m \end{pmatrix}.$$

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- Putting them in Dirac equation for u-spinors

$$\gamma^\mu p_\mu - m = \gamma^0 E_p - \boldsymbol{\gamma} \cdot \boldsymbol{p} - m = \begin{pmatrix} E_p - m & -\boldsymbol{\sigma} \cdot \boldsymbol{p} \\ \boldsymbol{\sigma} \cdot \boldsymbol{p} & -E_p - m \end{pmatrix}.$$

- If we write the u-spinors as

$$u \equiv \begin{pmatrix} \phi_t \\ \phi_b \end{pmatrix},$$

where ϕ_t and ϕ_b are top two and bottom two components respectively.

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Dirac-Pauli Representation

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- In the new notation the Dirac equation for u-spinors

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  Accordingly the u and v-spinors are written as

$$u_\pm(\boldsymbol{p}) = \sqrt{E_p + m}\begin{pmatrix}\chi_\pm\\ \frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{E_p+m}\chi_\pm\end{pmatrix}, \quad v_\pm(\boldsymbol{p}) = \sqrt{E_p + m}\begin{pmatrix}\frac{\boldsymbol{\sigma}\cdot\boldsymbol{p}}{E_p+m}\chi_\mp\\ \chi_\mp\end{pmatrix}.$$

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  The equation also can be simply found out from

$$\partial_{\mu}\left(\frac{\partial\mathscr{L}}{\partial\left(\partial_{\mu}\overline{\psi}\right)}\right) = \frac{\partial\mathscr{L}}{\partial\overline{\psi}}.$$

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- **Problem 6: Find the E-L equations for $\mathscr{L}'$.**
- **Problem 7: Find the γ_5 matrix in the Dirac-Pauli representation.**

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- What more is done???

References:

- A first book of Quantum Field Theory, CRC Press, **Amitabha Lahiri and Palash Baran Pal**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.